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Randomness of infinite sequences

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Problem: Can we extend the algorithmic definition of randomness to infinite sequences?

The link established by Andrei Kolmogorov between randomness and incompressibility could have put an end to a half-century quest for a definition of randomness. Extending the equation ‘randomness = incompressibility’ to infinite sequences didn’t prove straightforward, though.

An infinite series of fair coin tosses produces a random sequence. Conversely, how can we be sure that an infinite binary sequence was *not* produced by flipping a coin?

Per Martin-Löf (1966) defined tests to capture non-randomness. A random sequence must escape all of them. For instance, a sequence cannot be considered random if it has too many 0 at the beginning, or many more 0s than 1s, or a 1 at each even position, or either 01 or 10 at even positions.



Randomness tests

With the support of the Patrick and Lina Drahi Foundation

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if a string is close to incompressible, this means that it contains very few regularities and thus passes all frequency-based tests. Martin-Löf’s result shows that it also passes all other possible P-tests.

Conclusion: Infinitely-many tests can be used independently to define randomness, but most of them are not sufficient

Kolmogorov got the intuition that any failed test would made the sequence compressible. Generalizing this property is not immediate. Building on Kolmogorov’s work, Claus Schnorr and Leonid Levin found in 1973 that defining randomness for infinite sequences requires referring to prefix complexity K instead of plain complexity C .

If we call $x_{1:m}$ the beginning of sequence x up to index m , then **a series x is random** if it is incompressible, which means that any finite chunk of it is roughly as complex as its length.

Randomness of infinite binary sequences

$$(\exists c) (\forall m) K(x_{1:m}) > m - c$$

Randomness of Pi

1/1 point (ungraded)

Should π 's decimal development, as an infinite sequence, be regarded as random?

☐ Yes

☒ No

✓

Answer

Correct:

π 'decimals are by no means random, as they can be compressed down to a *finite*

(and quite limited!) size.

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✓ Correct (1/1 point)

Randomness generation

1/1 point (ungraded)

Can we generate a random (= incompressible) infinite sequence by keeping a computer running for ever?

☐ Yes

☒ No



Answer

Correct:

The answer is No. The program p used to generate the sequence s offers a summary of it, and so $K(s) < |p| + c$. As a consequence, such a sequence should be considered as highly compressible and not random at all. This is why computers can only generate what is called "pseudo-random" sequences. You might imagine to get truly random bits from an external 'oracle' (for instance a noisy resistor used as a register or a quantum device). That's OK, but the output is not reproducible and there is no way to be sure that what you get is really random. Note that living organisms seem to generate randomness this way, *e.g.* by allowing physical noisy processes to mutate DNA.

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✓ Correct (1/1 point)

Rational numbers 1

0/1 point (ungraded)

A rational number r is the quotient of two integers $r = p/q$. Can some rational numbers be considered random?

☐ Yes

☒ No



Answer

Incorrect:

A rational number can be considered as a pair of integers. If both of them are incompressible for plain Kolmogorov complexity (on a given machine), then the rational number should be regarded as random.

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✗ Incorrect (0/1 point)

Rational numbers 2

1/1 point (ungraded)

Consider the binary expansion of a rational number. If infinite, can it be regarded as a random sequence?

☐ Yes

☒ No

Answer

Correct:

No. As an infinite binary sequence, it is highly compressible , since the development of rational numbers is necessarily periodic at some point. Note that though some rational numbers may genuinely be regarded as random (as finite objects), their binary expansion, if infinite, is by no means a random infinite sequence.

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✔ Correct (1/1 point)



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